

Volunteer Event Management System (VEMS)

OutSystems Architecture Canvas

This document describes the layered architecture of the Volunteer Event Management System (VEMS), aligned with the OutSystems 11 Architecture Canvas principles. The architecture is structured into End-User, Core, Foundation, and External Systems layers.

1. End-User Layer (Applications)

Applications expose functionality to volunteers and admins. They consume only Core modules and the Design System.

- VEMS_App_VolunteerWeb – Event browsing, event details (with live weather), sign-up/unregister, My Events dashboard, feedback.
- VEMS_App_AdminWeb – Event and location management (CRUD), participant management, dashboards, reporting.
- VEMS_App_Mobile (optional) – Volunteer check-in (geolocation & radius validation), camera photo upload, offline queue & sync.

2. Core Layer (Domain & Orchestration)

Core modules encapsulate business logic and own entities. They consume Foundation services.

- VEMS_Core_Events – Entities: Event, Location, WeatherCache. Event creation, publishing, completion, weather orchestration.
- VEMS_Core_Participation – Entity: Registration. Volunteer sign-up, unregister, capacity guard, registration status.
- VEMS_Core_Feedback – Entity: Feedback. Feedback submission, summaries, rules for visibility.
- VEMS_Core_Attendance (optional) – Entity: Attendance. Check-in records, validation of check-in windows & geofence.
- VEMS_Core_Notifications – Entity: NotificationLog. Manage reminders and follow-ups, EventReminderTimer, logs.
- VEMS_Proc_PostEvent (optional BPT) – Triggered on event completion, generates reports, sends follow-ups, aggregates feedback.

3. Foundation Layer (Cross-Cutting Services)

Foundation modules provide reusable technical services and integrate with external systems. They do not own domain entities.

- VEMS_Fnd_DesignSystem – UI theme, patterns, toasts, empty states.
- VEMS_Fnd_Auth – User roles, profile bootstrap, OutSystems Users integration.

- VEMS_Fnd_WeatherProvider – REST connector to OpenWeather API, caching, errors, DTOs.
- VEMS_Fnd_Messaging – Email, SMS, push notification wrappers.
- VEMS_Fnd_GeoUtils (Extension) – Distance calculations (Haversine), geofence validation.
- VEMS_Fnd_Logging – Structured logging, exception helpers, feature flags.

4. External Systems

Integrated only via Foundation modules:

- OpenWeather – REST API for live weather data.
- Email/SMS Providers – Outbound transactional communication.
- Maps/Geo – Map tiles, mobile geolocation plugin support.

5. Composition Rules

- Top-down dependencies only: End-User → Core → Foundation → External Systems.
- No lateral dependencies between Core modules.
- One entity owner per Core module.
- Expose minimal Public APIs (DTOs, actions).
- Idempotent timers (reminders sent once only).
- Site Properties for API keys, thresholds, system parameters.

6. Architecture Benefits

- Scalability: Business capabilities isolated in Core modules.
- Maintainability: Clear separation reduces coupling.
- Extensibility: Optional modules (Mobile, BPT, Offline, Extension) can be added easily.
- Compliance: Aligns with OutSystems 11 Architecture Canvas best practices.